Terraform Assignment

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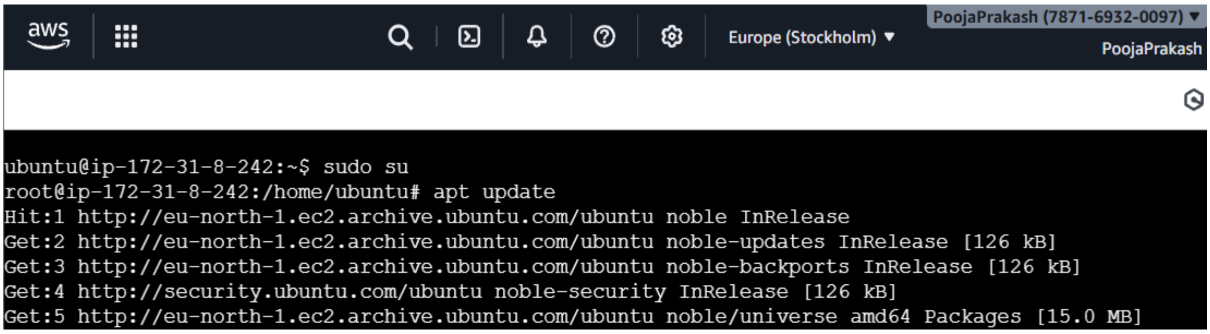
1. L1 - Provision AWS EC2 Instance along with VPC, Subnet, Internet Gateway, Route-Table, Route Table Association, Security Group

- Launch an EC2 instance, connect to it, and install Terraform on the EC2 machine.




```
root@ip-172-31-8-242:/home/ubuntu# terraform --version
Terraform v1.14.4
on linux amd64
```

- Create a folder named **terraform-assignment** to work with Terraform.
- Configure the AWS account by creating a configuration.tf file and providing the Access Key, Secret Key, and region.

```
root@ip-172-31-8-242:/home/ubuntu# mkdir terraform-assignment
root@ip-172-31-8-242:/home/ubuntu# cd terraform-assignment
root@ip-172-31-8-242:/home/ubuntu/terraform-assignment# vi configuration.tf
root@ip-172-31-8-242:/home/ubuntu/terraform-assignment#
```

```
provider "aws" {
  region     = "eu-north-1"
  access_key = "AKIA3ORXSTS[redacted]"
  secret_key = "UoM2mptMTaSsrGu9tL+qpb[redacted]"
}
~
~
~
~
```

- Create the main.tf script file to provision:
  - VPC
  - Subnet
  - Internet Gateway
  - Route Table
  - Route Table Association
  - Security Group
  - EC2 instance

```
root@ip-172-31-8-242:/home/ubuntu/terraform-assignment# ls
configuration.tf
root@ip-172-31-8-242:/home/ubuntu/terraform-assignment# vi main.tf
root@ip-172-31-8-242:/home/ubuntu/terraform-assignment# cat main.tf
#1. VPC
resource "aws_vpc" "main_vpc" {
  cidr_block  = "10.0.0.0/16"

  tags = {
    Name = "Assign-VPC"
  }
}
```

```
#2. Subnet
resource "aws_subnet" "pub_subnet" {
  vpc_id     = aws_vpc.main_vpc.id
  cidr_block = "10.0.1.0/24"
  availability_zone = "eu-north-1a"
  map_public_ip_on_launch = true

  tags = {
    Name = "Assign-Subnet"
  }
}

#3. Internet Gateway
resource "aws_internet_gateway" "igw" {
  vpc_id = aws_vpc.main_vpc.id

  tags = {
    Name = "Assign-IGW"
  }
}
```

```
#4. Route Table
resource "aws_route_table" "route_table" {
  vpc_id = aws_vpc.main_vpc.id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.igw.id
  }

  tags = {
    Name = "Assign-RT"
  }
}

#5. Route Table Association
resource "aws_route_table_association" "rt_assoc" {
  subnet_id      = aws_subnet.pub_subnet.id
  route_table_id = aws_route_table.route_table.id
}
```

```
#6. Security Group
resource "aws_security_group" "sg" {
  name        = "Assign-sg"
  description = "Allow SSH and app access"
  vpc_id      = aws_vpc.main_vpc.id

  # SSH access
  ingress {
    description = "SSH"
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  # Allow all outbound traffic
  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
```

```
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }

  tags = {
    Name = "Assign-sg"
  }
}

#7. EC2 Instance
resource "aws_instance" "ec2_instance" {
  ami                         = "ami-073130f74f5ffb161"
  instance_type               = "t3.micro"
  subnet_id                   = aws_subnet.pub_subnet.id
  vpc_security_group_ids      = [aws_security_group.sg.id]
  associate_public_ip_address = true
  key_name                    = "linux1"

  tags = {
    Name = "Assign-EC2"
  }
}
root@ip-172-31-8-242:/home/ubuntu/terraform-assignment#
```

- Run the *terraform init* command to initialize Terraform in the project directory.

```
root@ip-172-31-8-242:/home/ubuntu/terraform-assignment# terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.31.0...
- Installed hashicorp/aws v6.31.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
root@ip-172-31-8-242:/home/ubuntu/terraform-assignment#
```

- Run the *terraform plan* command to generate an execution plan that shows what Terraform will create when the *terraform apply* command is executed.

```
root@ip-172-31-8-242:/home/ubuntu/terraform-assignment# terraform plan

Terraform used the selected providers to generate the following execution plan. Resource
actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.ec2_instance will be created
  + resource "aws_instance" "ec2_instance" {
      + ami                                  = "ami-073130f74f5ffb161"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = true
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
```

```
Plan: 7 to add, 0 to change, 0 to destroy.

─

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to
take exactly these actions if you run "terraform apply" now.
root@ip-172-31-8-242:/home/ubuntu/terraform-assignment#
```

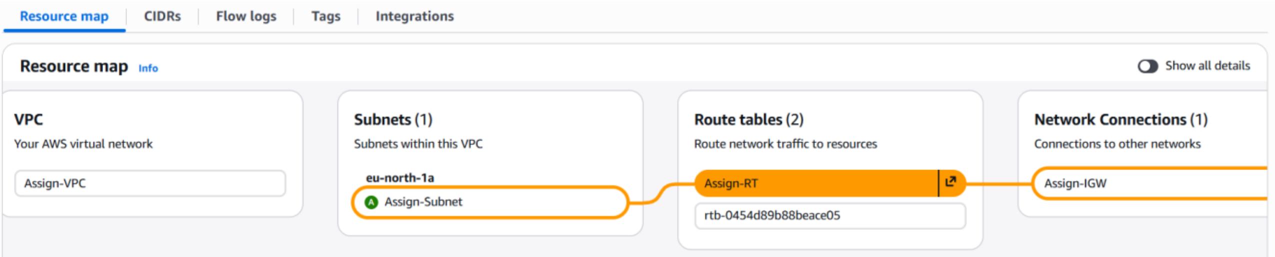
- Finally, run the *terraform apply* command to create the required infrastructure defined in the .tf files.

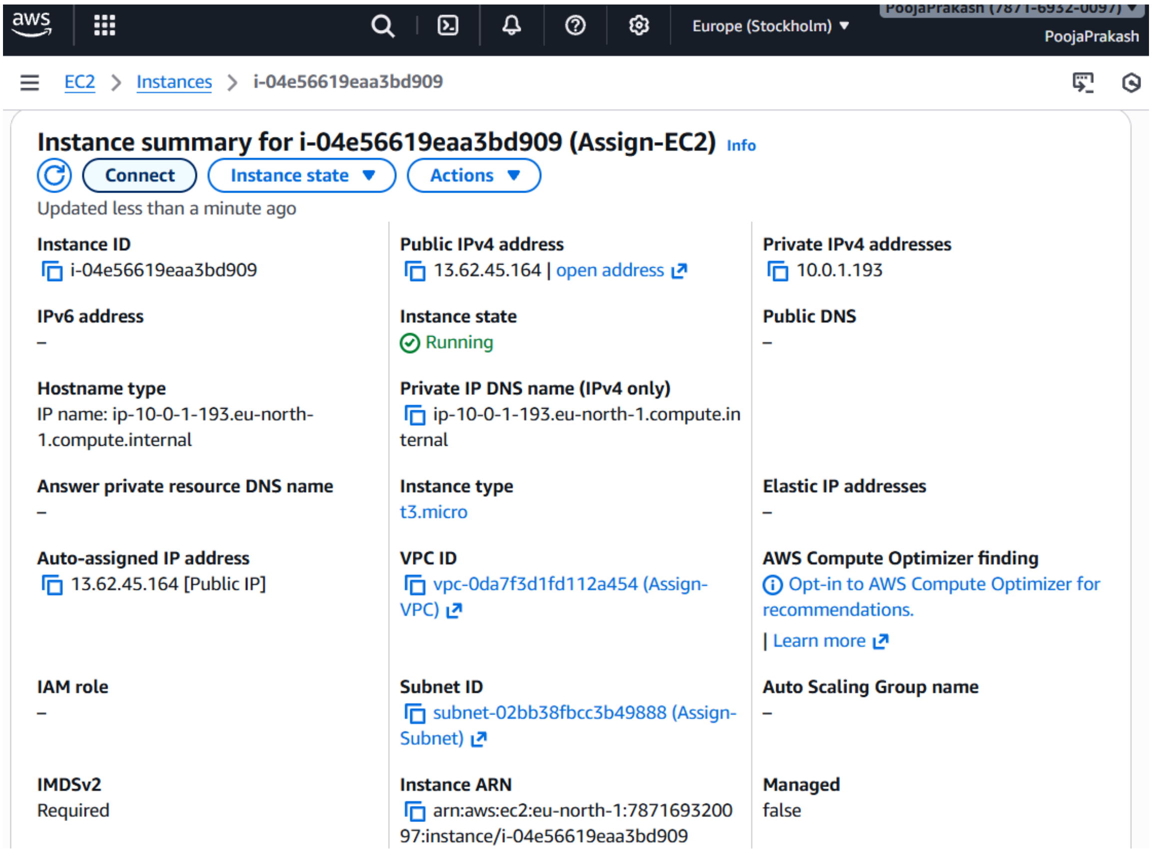
```
root@ip-172-31-8-242:/home/ubuntu/terraform-assignment# terraform apply
```

```
  Enter a value: yes

aws_vpc.main_vpc: Creating...
aws_vpc.main_vpc: Creation complete after 1s [id=vpc-0da7f3d1fd112a454]
aws_subnet.pub_subnet: Creating...
aws_internet_gateway.igw: Creating...
aws_security_group.sg: Creating...
aws_internet_gateway.igw: Creation complete after 1s [id=igw-0358aae6671b602f8]
aws_route_table.route_table: Creating...
aws_route_table.route_table: Creation complete after 1s [id=rtb-0bee0df1a6a46b019]
aws_security_group.sg: Creation complete after 3s [id=sg-0d8597d8b3817f8a6]
aws_subnet.pub_subnet: Still creating... [00m10s elapsed]
aws_subnet.pub_subnet: Creation complete after 11s [id=subnet-02bb38fbcc3b49888]
aws_route_table_association.rt_assoc: Creating...
aws_instance.ec2_instance: Creating...
aws_route_table_association.rt_assoc: Creation complete after 1s [id=rtbassoc-0f9d33ffba1
c30257]
aws_instance.ec2_instance: Still creating... [00m10s elapsed]
aws_instance.ec2_instance: Creation complete after 13s [id=i-04e56619eaa3bd909]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.
root@ip-172-31-8-242:/home/ubuntu/terraform-assignment#
```


Resource map
CIDRs
Flow logs
Tags
Integrations
Resource map Info
Show all details
VPC
Your AWS virtual network
Assign-VPC
Subnets (1)
Subnets within this VPC
eu-north-1a
Assign-Subnet
Route tables (2)
Route network traffic to resources
Assign-RT
rtb-0454d89b88beace05
Network Connections (1)
Connections to other networks
Assign-IGW


aws
Europe (Stockholm)
PoojaPrakash
EC2
Instances
i-04e56619eaa3bd909
Instance summary for i-04e56619eaa3bd909 (Assign-EC2) Info
Connect
Instance state
Actions
Updated less than a minute ago
Instance ID
i-04e56619eaa3bd909
Public IPv4 address
13.62.45.164 | open address
Private IPv4 addresses
10.0.1.193
IPv6 address
–
Instance state
Running
Public DNS
–
Hostname type
IP name: ip-10-0-1-193.eu-north-1.compute.internal
Private IP DNS name (IPv4 only)
ip-10-0-1-193.eu-north-1.compute.internal
Answer private resource DNS name
–
Instance type
t3.micro
Elastic IP addresses
–
Auto-assigned IP address
13.62.45.164 [Public IP]
VPC ID
vpc-0da7f3d1fd112a454 (Assign-VPC)
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
| Learn more
IAM role
–
Subnet ID
subnet-02bb38fbcc3b49888 (Assign-Subnet)
Auto Scaling Group name
–
IMDSv2
Required
Instance ARN
arn:aws:ec2:eu-north-1:787169320097:instance/i-04e56619eaa3bd909
Managed
false